Vector Electronics

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The purpose of this article is to describe an attempt to solve electronic circuits made of non-linear devices like diodes, bipolar transistors, JFETs, MOSFETs, etc.

A common approach is to set the circuit at some operating point and wiggle it around with the input signal. For small variations, even the most non-linear device can be approximated by a linear model. The *Small Signal Analysis*, as it is called, is a classical framework for that.

However, some circuits are trickier and make one or more nonlinear devices operate way beyond the small signal hypothesis.

The class AB (push/pull) amplifier is one example: a transistor in the pair goes from cutoff to linear region as the input rises. The other transistor does the opposite. Meanwhile, the output follows the input nicely (the circuit as a whole surprisingly acts as if everything were linear.)

To fully analyse this, one could try to use a full model of the transistor (exponential curve) however:

- the equations usually have no closed-form expression (equations combining exp() and polynomials)
- it doesn't give much insight on *how* the thing works.

The Piecewise Linear Model

A nonlinear continuous function can always be approximated by a piecewise affine (linear) function[1], provided the model is refined and enough control points are introduced. Thus, when analyzing a circuit, one can quite reasonably assume, as a first approximation, that the characteristic of a nonlinear component is in fact piecewise linear.
The parameters of the linear relationship (slope and offset) may not yet be known for the present operating point, but they must necessarily correspond to one of the (slope, offset) pairs among those defined in the proposed piecewise affine model.

[1] In other words, the set of piecewise linear functions is dense in the space of continuous real-valued functions

The task then reduces to determining which pair applies at a given operating point. The existence of such a pair is ”reassured” by the fact that every circuit necessarily has *at least* one operating point. It is reasonable to assume that this solution is unique.[2]

For most practical cases, the overall result is that every component can be treated as linear, thereby enabling the full use of linear analysis techniques (superposition principle, etc.) without any loss of generality.

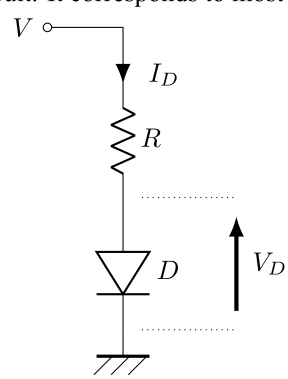
Notice that, since any linear transformation to a piecewise linear function is still a piecewise linear function, the very concrete consequence is that the piecewise-affine property of a nonlinear component **propagates** to all the properties of the electronic circuit in which it is embedded.

In this sense, we speak in this paper of a *Vector Analysis* of the circuit. If one views a piecewise-affine function as an element of a vector space (which it is) then the circuit's properties—whatever they may be: V_{OUT} vs V_{IN}, I/V characteristic, transmittance, input impedance, etc.—are likewise contained within that same vector space.

It may therefore be conceivable to interpret the action of a circuit on a nonlinear component as analogous to a matrix acting on a vector. The idea that a circuit might have a kind of “matrix analogue” is perhaps worth exploring further.

Example 1: diode and resistor I

One of the first case study for a diode in a semiconductor lecture is the resistor-diode circuit. It corresponds to most of the LED polarisation circuits.



Even though this is probably the simplest circuit involving a diode, the exponential behaviour of the I/V curve takes a bit of method to solve the circuit.

Here are some common approaches:

- Assume $V_D = 0.7V$ (or a few Volts for LEDs), determine R for a target current (say $5mA$) →Done! It's precise enough for most applications.

- Iterative solving:

[2]Even though certain ”pathological” circuits (e.g. with feedback) may admit multiple possible operating points (stable or not). We will return to this point later.

1. Assume a *guess* voltage $V_D = 0.7V$ accross the diode (no need to be accurate)
2. Determine the current I_D using the diode $I{-}V$ model: $I_D = I_S \left(\exp\left(\frac{V_D}{V_T}\right) - 1 \right)$
3. calculate the new V_D using $V_D = V - RI_D$
4. go back to 2. until V_D and I_D eventually settle.

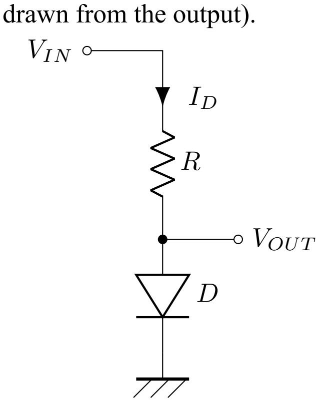
Convergence rate is insanely fast and the result gets accurate to a few digits after only 2 or 3 iteration. However, this method is more of a theoretical fact. It might be useful in a simulator, but for practical situations it won't bring much more precision. The natural dispersion in I/V curves within the same batch of diodes, not even mentionning the accuracy of the model itself lowers the benefit of this method.

- Solve using Lambert's W function This function pops naturally when solving equations like $exp(x) = ax + b$. Though the conclusions are similar to the iterative method.

Example 2: diode and resistor II

This is a slight variation of the previous diode circuit.

We now have a varying voltage source as input of the system, the diode voltage is the output. We want to see how the output varies with the input (assuming no current is drawn from the output).



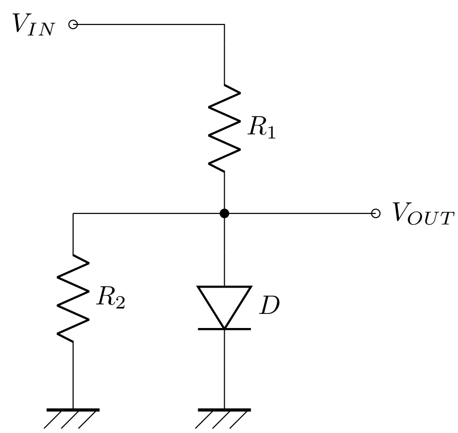
For $V_{IN} > 0.7V$ a first good approximation is $V_{OUT} \simeq 0.7V$ regardless of the input voltage V_{IN}.

It makes sense: for an increasing V_{IN}, the diode voltage cannot follow the input and rise as much: the exponential law would kick in and cause a massive increase in current, thereby reducing the diode voltage (through the resistor), opposing the initial voltage increase.

In a way, the diode voltage **does not** want to change.

Like before, if we just assume $V_D = 0.7V$ This is where the Piecewise Linear model can become handy.

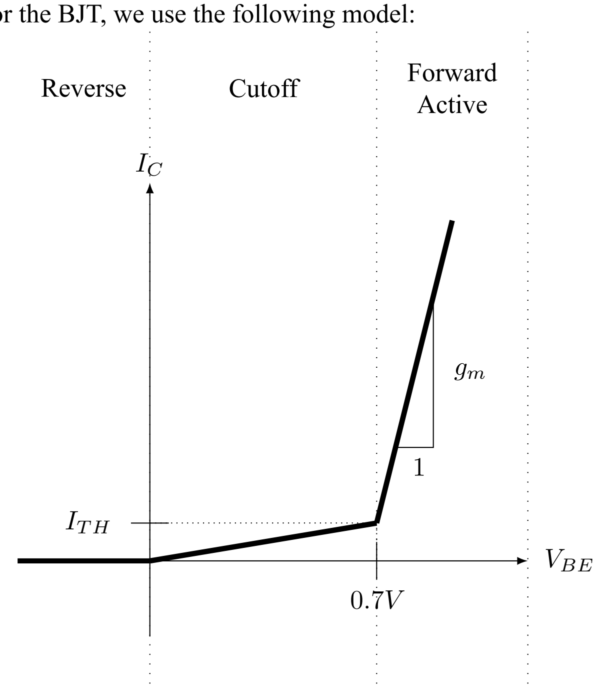
Example 3: diode and resistor III



Example 4: the Common Collector (or *Emitter Follower*)

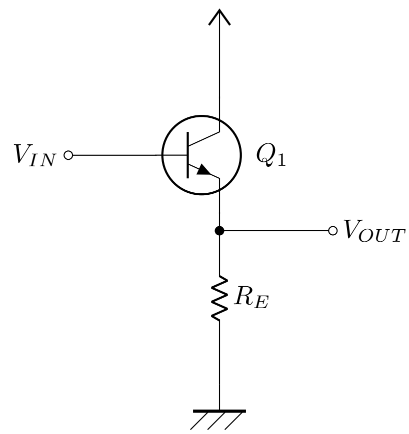
Derivation

For the BJT, we use the following model:



As we will see, for the common collector, the actual values for g_m and I_{TH} do not matter that much. However, this is not true in the general case. We will illustrate that in later examples.

The Common Collector circuit:



So just like we did with the diode previously: we assume a linear model $I_C = f(V_{BE}) = aV_{BE} + b$ with a and b depending on the operating point (Reverse, Cutoff, Forward Active)

$$V_{OUT} = R_E f(V_{IN} - V_{OUT}) = aR_E\,(V_{IN} - V_{OUT}) + b$$

Rearranging for V_{OUT}:

$$V_{OUT} = \frac{aR_E}{1 + aR_E} V_{IN} + \frac{b}{1 + aR_E}$$

A BJT is an op amp in disguise.

We've just seen that a BJT in the common emitter arrangement is basically always 'on'. It's the very high transconductance past the $V_{BE} = 0.7V$ that makes all the system work.
Applying a very high gain to a voltage difference reminds us of op amps. Under the hood, an op amp is voltage difference amplifier with a massive gain ($> 10^6$)
The high gain is rarely used as such because it is not very predictable, controlled and uniform in frequency. We much prefer to use this 'unreliable' gain in a feedback loop. It reduces the amplification, but it is more robust to part to part fluctuation

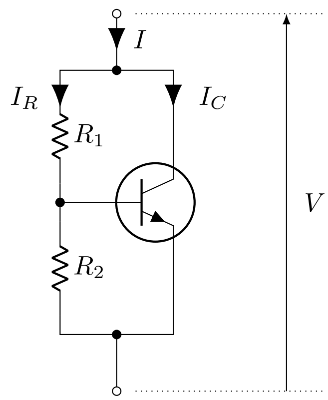
Example 5: the Rubber Diode

Description

In a Class AB Amplifier, we need to 'pre-heat' the NPN and PNP transistors of the final stage. This allows a quiescent current to flow at rest and maintain both transistor in active region. It is mandatory to get rid of the *crossover distortion*.

The polarisation is done usually with 2 diodes in series. But for more flexibility, one can also use the *Rubber Diode* below.

It is quite common to have a potentiometer for the resistors R_1 and R_2 to fine-tune the voltage V.



Derivation

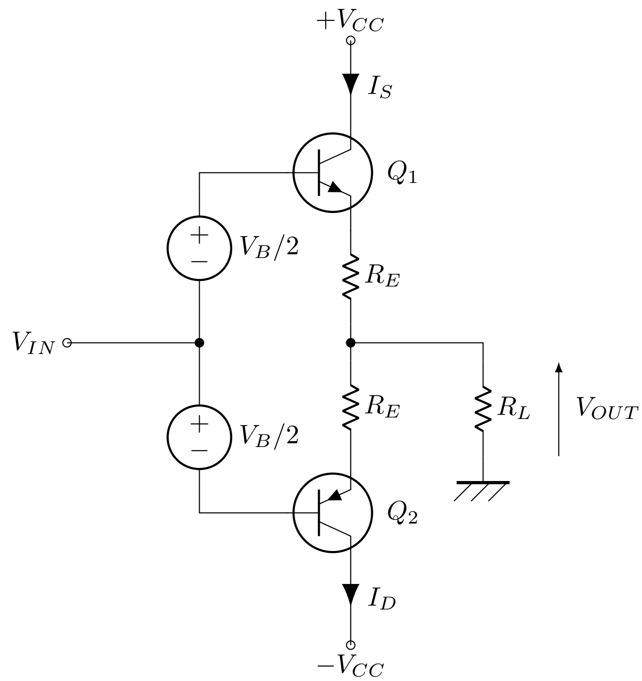
As always, it is safe to neglect the base current.

$$\begin{aligned} I &= I_R + I_C \\ &= \frac{V}{R_1 + R_2} + f\left(\frac{R_2}{R_1 + R_2} V\right) \\ &= \frac{V}{R_1 + R_2} + a\frac{R_2}{R_1 + R_2} V + b \\ &= \frac{1 + aR_2}{R_1 + R_2} V + b \end{aligned} \tag{1}$$

The solution is more straightforward as there is no feedback involved here. Observations:

- starts to act as a diode when $\frac{R_2}{R_1+R_2} \simeq 0.7V$
- when active, the new transconductance becomes $\hat{g}_m = \frac{1+g_m R_2}{R_1+R_2}$

Example 6: class AB amplifier



Derivation Assuming that $I_C = f_S(V_{BE})$ for the NPN and $I_C = f_D(V_{EB})$ for the PNP we get the following:

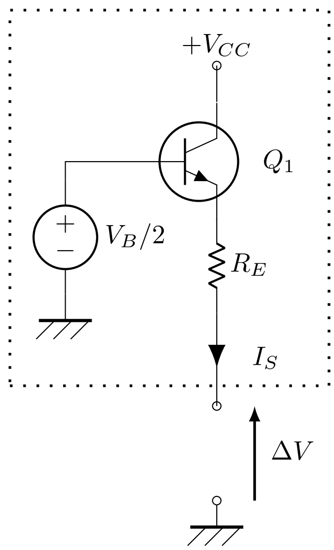
$$\begin{cases} I_S = f_S\left(v_{IN} - v_{OUT} + \frac{v_B}{2} - R_E I_S\right) \\ I_D = f_D\left(-(v_{OUT} - v_{IN}) + \frac{v_B}{2} - R_E I_D\right) \\ V_{OUT} = R_L(I_S - I_D) \end{cases}$$

This time, the equations are trickier: the key variables I_S, I_D and V_{OUT} are all intertwined. We cannot carry out the analysis the way we did in the previous sections.

Let's try to analyse the system as 2 independant subsystems.

Class AB subsystem: the current source

If we split the full schematic in half and set the input voltage to a constant, we can simplify the current source of the circuit to a much simpler block:



Using the usual voltage and currents laws gives the following source current:

$$I_S = f_S\left(-\Delta V + \frac{v_B}{2} - R_E I_S\right)$$

Under a linear model assumption $f_S(V) = aV + b$, we can write that:

$$I_S = a\left(-\Delta V + \frac{v_B}{2} - R_E I_S\right) + b$$

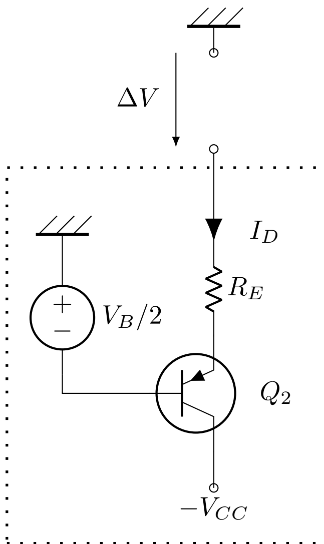
Finally, after a bit of algebra:

$$\boxed{I_S = -\frac{1}{R_E}\frac{aR_E}{1+aR_E}\Delta V + \frac{a\frac{V_B}{2}+b}{1+aR_E} = \hat{f}_S(\Delta V)}$$

If we apply this equation directly on the BJT model described earlier, we can now plot the full curve I_S vs ΔV and see how this device reacts as we change the control voltage ΔV.

Class AB subsystem: the current drain

Similarly, we extract the lower half of the circuit:



The equation looks like the one we got for the current source version:

$$I_D = f_D\left(\Delta V + \frac{v_B}{2} - R_E I_D\right)$$

And: $\boxed{I_D = \frac{1}{R_E}\frac{aR_E}{1+aR_E}\Delta V + \frac{a\frac{V_B}{2}+b}{1+aR_E} = \hat{f}_D(\Delta V)}$

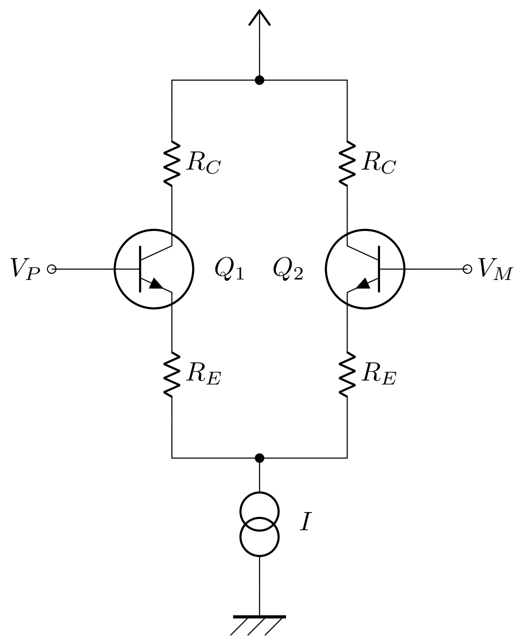
Class AB subsystem: summary

Let's see now how we can combine the subsystems to solve for the class AB entirely. If we treat these subsystems as a *virtual device* (nonlinear) the original system of equation becomes much simpler:

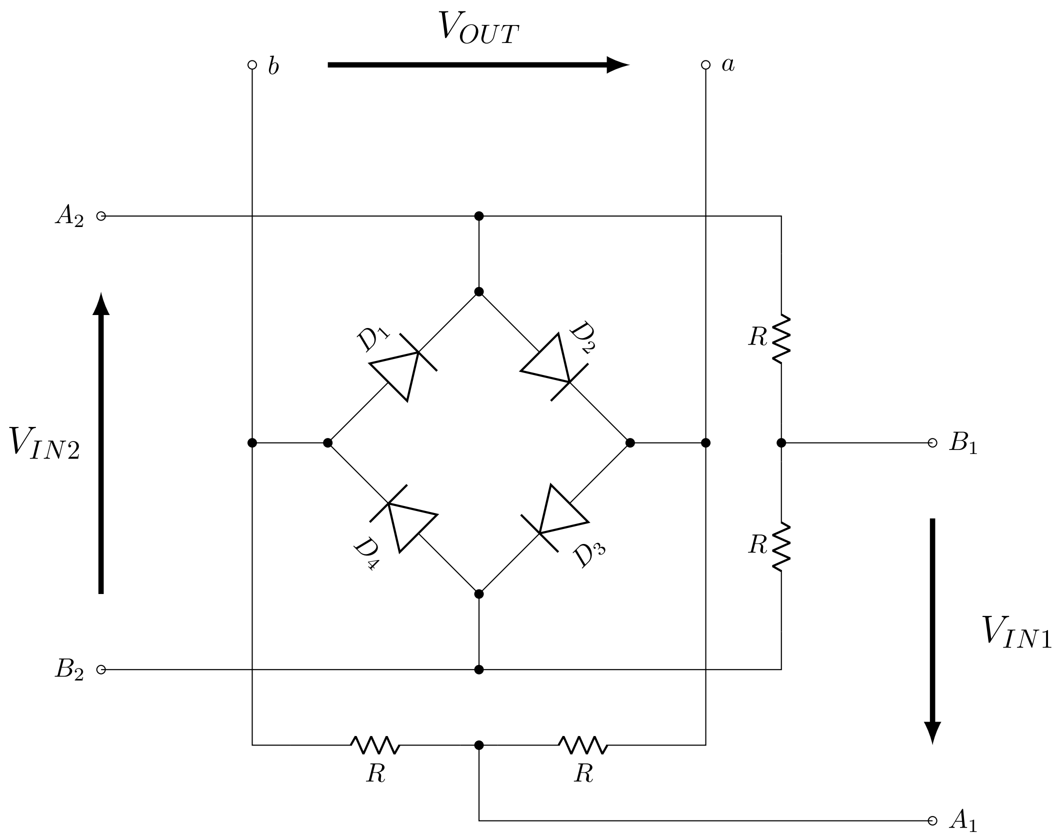
$$V_{OUT} = R_L\left(\hat{f}_S\left(v_{IN} - v_{OUT}\right) - \hat{f}_D\left(v_{OUT} - v_{IN}\right)\right)$$

Notice that now, I_S and I_D are no longer defined recursively, and also the v_B and R_E terms are integrated in the nonlinear functions $\hat{f}_S$ and $\hat{f}_D$

Example 7: the Differential Pair



Example 8: the Ring Modulator



Example 9: the Gilbert Cell (analog multiplier)

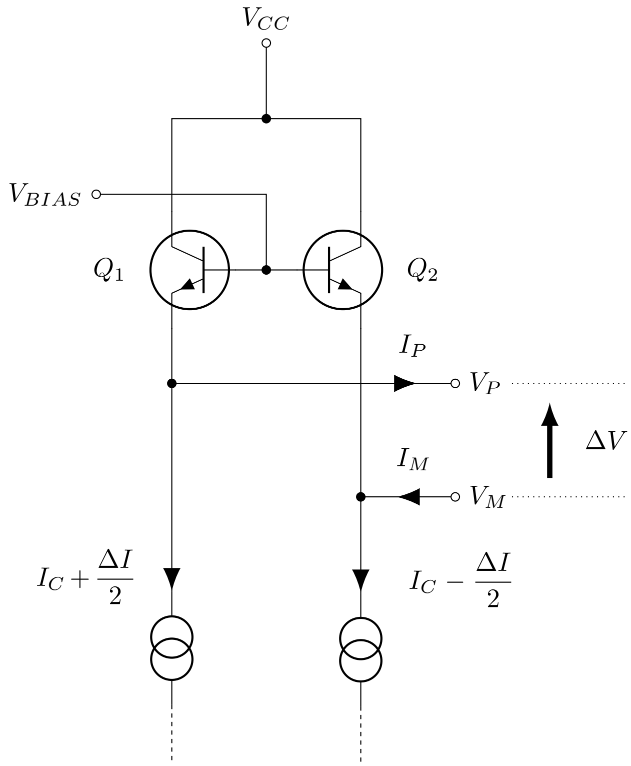
That's a tough one, be patient...

Example 10: the Moog Ladder Filter I

Description

The Moog Ladder Filter is a parametric low-pass filter used in audio applications. Its cutoff frequency can be freely adjusted using a control current, which makes it a very unique kind of filter. It gained its popularity and fame in the Moog Synthesizers, as early as the seventies.

The term *Ladder Filter* comes from the way the schematic looks: its very much *ladder-like*. In this section, we don't analyse the full schematic, only the 'top' part of the filter:



To simplify, we assume here that the bottom section of the ladder is a balanced current source: two branches, a 'common mode' **control current** I_C to which a 'differential mode' **signal current** $\frac{\Delta I}{2}$ is superimposed.

Such source is easily done with the Differential Pair studied earlier.

Derivation

As usual we denote the collector current I_C as $I_C = f(V_{BE})$ and get:

$$\begin{cases} I_P = f(V_{BIAS} - V_P) - I_C - \frac{\Delta I_C}{2} \\ I_M = -f(V_{BIAS} - V_M) + I_C - \frac{\Delta I_C}{2} \end{cases}$$

With the piecewise linear assumption:

$$\begin{cases} I_P = -aV_P - I_C - \dfrac{\Delta I}{2} + aV_{BIAS} + b \\ I_M = aV_M + I_C - \dfrac{\Delta I}{2} - aV_{BIAS} - b \end{cases}$$

Once the output is loaded, $I_P = I_M \triangleq I$, $V_P - V_M \triangleq \Delta V$ and summing the 2 equations gives:

$$-a\Delta V - \Delta I = 2I$$

Put differently:

$$\Delta V = -\frac{I + \Delta I}{a}$$

If the load is a simple impedance Z, then $I = \frac{\Delta V}{Z}$ and it becomes:

$$\Delta V = -\frac{Z}{2 + aZ}\Delta I$$

If we write that $a \triangleq \frac{1}{R_C}$ (the *dynamic* resistance), we get this rather *nice* expression:

$$\Delta V = -\frac{1}{2}\left[\frac{(2R_C)Z}{(2R_C) + Z}\right]\Delta I$$

An observer looking at the impedance Z would actually see a parallel association of the impedance Z with some resistance R_C, with some $-1/2$ gain. Recall that ΔI is a *current* replica of the input signal to be filtered. Now, it is key to remember that a is the transconductance of the BJT in the model.

In the previous circuits, just assuming that a is much greater than 1 was enough. Here, we need some more hypothesis.

Comparison without active loading

To fully understand this result, we now study what would happen without the transistors.

The Resistive load case

With $Z = R$ we simply get:

$$\Delta V = -\frac{1}{2}\left[\frac{(2R_C)R}{(2R_C) + R}\right]\Delta I$$

From the output perspective, we see a parallel resistor association: $2R_C$ and R, with some gain $-1/2$.

The Capacitive load case

For a capacitive load, $Z = \frac{1}{jC\omega}$. We write $a = \frac{1}{R}$ thus:

$$\Delta V = R\frac{-1}{1 + 2jRC\omega}\Delta I$$

Now the key element to understand how this can be a parametric filter is to go back to the definition of a: a is a function of the control current I_C.

In the previous examples we didn't care much about the transconductance a.